Aviroop Pal

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Summary

AI Engineer with hands-on experience designing and shipping real-time, production-ready AI systems. Skilled at building end-to-end pipelines that ingest and process multimodal data (audio, vision, and sensor streams), integrate large language models, and expose them through scalable, containerized backends using Python, PyTorch, TensorFlow, FastAPI, and Docker.

Technical Skills

- **Languages & Frameworks:** Python, Go, C, JavaScript/TypeScript, FastAPI, Django, Flask, React, Node.js, Express.js, Next.js, Streamlit
- **Backend & APIs:** RESTful API design, WebSockets, SQLAlchemy, SQLite, PostgreSQL, Redis, Pydantic, asynchronous system architecture
- **Machine Learning & AI:** PyTorch, TensorFlow, Keras, scikit-learn, CNN, LSTM, YOLO, BERT, Mistral, LoRA, Transformers, LangChain, Retrieval-Augmented Generation (RAG)
- **Data Science:** Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook, data analysis and visualization
- **NLP:** NLTK, Transformers, BERT, LangChain, RAG pipelines
- **Computer Vision:** OpenCV, CNN, YOLO
- **DevOps & MLOps:** Docker, Git, GitHub Actions (CI/CD), Poetry, MLflow, Weights & Biases, Loguru
- **Languages Spoken:** English, Hindi, Bengali

Experience

AirDawgLabs Jul 2026 – Oct 2026
AI Systems Evaluation Intern *Remote*

- Evaluated LLMs and generative AI systems using structured testing frameworks to benchmark reliability, safety, and output quality against defined guidelines.
- Conducted prompt engineering, qualitative reviews, and analytical evaluations of model performance; contributed to validation pipelines and technical documentation.
- Used Cursor Pro and LLM-assisted tooling to accelerate analysis, structured data annotation, and testing.

NeoCortex LLC Apr 2026 – May 2026
AI/ML Intern *Remote*

- Contributed to AI-powered Chrome extensions focused on automation and productivity workflows, integrating browser APIs with AI-driven backend systems and REST APIs.
- Assisted in designing scalable, secure extension architectures for production deployment.
- Participated in testing, debugging, and deployment workflows in a collaborative remote setting.

BaseApp Dec 2025 – Feb 2026
AI/ML Intern *Remote*

- Built a real-time Voice-to-Voice (V2V) AI system with sub-second latency, integrating Speech-to-Text, LLMs, and Text-to-Speech into one pipeline.
- Implemented a WebSocket-based streaming API for bidirectional audio/text communication and real-time response generation.
- Designed a scalable asynchronous backend with Redis-based session management and Dockerized deployment.
- Hardened the system for production with authentication, DoS protection, memory optimization, and GPU acceleration.

Proption Fintech Private Limited Nov 2025 – Feb 2026
AI/ML Intern *Remote*

- Built internal company tools in Go to streamline operations.
- Designed and implemented financial algorithms, including a Supertrend indicator for NIFTY, to support data-driven decision-making.
- Contributed to backend and frontend development and to data collection/preprocessing that supported algorithm evaluation.

TherapyU, BIRAC Cell – Adamas University Jan 2025 – Feb 2025
AI/ML Intern *West Bengal*

- Developed and implemented production-ready AI/ML algorithms to enhance a web application's functionality, collaborating with UI/UX and development teams.

- Conducted data analysis, preprocessing, and model optimization for performance and scalability; rapidly prototyped AI solutions and produced technical documentation.

Projects

Vanta – Self-Hosted Deep Research API *FastAPI, PostgreSQL, Redis, Docker*

- Built a self-hosted, privacy-first Research-as-a-Service platform automating multi-round research workflows: query planning, web search, content extraction, fact synthesis, and cited report generation.
- Architected a scalable, asynchronous, queue-based modular monolith with database persistence, webhook integrations, report exports, source tracking, and conversational report querying.
- Technologies: FastAPI, Python 3.11, PostgreSQL, Redis, SQLAlchemy, Alembic, ARQ, Docker, SearXNG, OpenAI, Anthropic, Gemini, OpenRouter.

WatchNT – Privacy-First Meeting Assistant *FastAPI, React, TypeScript, Chrome Ext.*

- Built an open-source, privacy-first meeting assistant capturing tab audio and live captions across Google Meet, Zoom Web, and Microsoft Teams Web to automate transcription and intelligence extraction.
- Architected a containerized FastAPI backend and Manifest V3 Chromium extension with local speech-to-text (Faster-Whisper), multi-stage parallel prompt pipelines using Pydantic JSON schemas, local SQLite persistence, and instant Markdown/CSV exports.
- Engineered a Bring-Your-Own-Keys (BYOK) model orchestration layer supporting fully offline inference via Ollama alongside Groq, OpenAI, Gemini, and OpenRouter.
- Technologies: Python 3.10, FastAPI, React 19, TypeScript, Vite, Tailwind CSS, Manifest V3 (CRXJS), Faster-Whisper, FFmpeg, Ollama, Pydantic v2, SQLAlchemy, SQLite, Docker.

NeuroSleepNet v1.0 – Memory Infrastructure for Small Language Models *PyTorch, LangChain*

- Developed a memory-augmented learning framework to mitigate catastrophic forgetting in AI agents, implementing latent replay buffers and automatic task-boundary detection.
- Designed LLM sidecar memory injection with monitoring dashboards for memory-retention analysis.

Real-Time Voice AI System *Python, FastAPI, Redis, Docker, WebSockets*

- Engineered a low-latency, conversational Voice-to-Voice AI platform integrating STT, LLM inference, and TTS into a scalable streaming architecture with asynchronous APIs and optimized GPU inference.

Federated Learning & Differential Privacy Research *PyTorch, TensorFlow*

- Researching privacy-preserving federated learning architectures and differential privacy techniques that maintain model accuracy across distributed nodes.
- Building scalable distributed training pipelines with CI/CD integration.

Education

Adamas University Expected 2027

B.Tech, Computer Science and Engineering *Kolkata, West Bengal*

- GPA: 8.0/10

Certifications

- Generative AI Fundamentals – Google
- Prepare Data for ML APIs on Google Cloud – DeepLearning.AI
- Create and Manage Cloud Resources – Google
- Enterprise Design Thinking – IBM
- Build a Secure Google Cloud Network – Google